

ONLINENOTEPAD QA AUTOMATION FRAMEWORK

Non-Technical User Guide & Test Specification for SQA Teams

1. How this Tool Helps SQA Teams

- **Eliminates Manual Regressions:** Automatically executes 19 complex scenarios in a visible browser.
- **Cross-Browser Verification:** Supports running tests in Google Chrome, Mozilla Firefox, and Microsoft Edge.
- **Evidence-Backed Bug Reports:** Captures screenshots, console logs, network activity, and stack traces on failures.
- **Dynamic Reports:** Generates Word (.docx) and PDF (.pdf) reports detailing exact pass/fail reasons.

2. First-Time Setup

Before running the automation suite, ensure you have Node.js installed on your system. Open your command line (terminal) in this folder, and run:

```
npm install
npx playwright install
```

3. How to Run Automated Tests

To start the interactive program, run the following command:

```
npm run test-cli
```

The tool will ask you to:

1. Select Browser (Chrome, Firefox, or Edge)
2. Select the Test Case (from the 19 registered test cases)
3. Select Report Format (Word, PDF, or Both)
4. Input Credentials (if authentication is required by the test)

Note: For tests requiring login, you may see a CAPTCHA challenge in the browser. Since CAPTCHA is designed to block automation, you will need to manually solve the challenge inside the opened browser window to let the test proceed.

4. Descriptions of the 19 Automated Test Cases

Guest User Scenarios (No Login Required)

- **1. Guest Note Persistence:** Verifies that guest notes are saved locally in the browser and remain available when you refresh the page.
- **10. Local Image Upload:** Verifies you can upload a local image file into the note editor and that it displays correctly.
- **11. Third Party Image Validation:** Verifies inserting a link to a remote image renders correctly without errors.

Account Registration & Login Scenarios

- **2. Guest Note Assignment After Login:** Verifies that guest notes are automatically assigned to the user account on sign-in.
- **4. Multiple Guest Notes Assignment:** Verifies multiple guest notes are successfully transferred to the user account on sign-in.
- **13. Account Switching:** Logs in User A, creates note, logs out, and signs in User B, confirming that User A's notes are isolated and hidden.

Security & Note Ownership

- **3. Ownership Validation:** Confirms that if guest notes are assigned to Account A, Account B cannot view or

access them under any circumstances.

Server Synchronization & Advanced Behaviors

- **5. Sync Status Validation:** Validates notes transition from "unsynced" (local storage) to "synced" (saved to cloud) once saved.
- **6. Sync Trigger On Open:** Confirms opening a note refreshes the sync status and updates modification times.
- **7. Cross Browser Sync:** Logs in same account in two browser instances, verifying notes synced in A immediately appear in B.
- **8. Placeholder Notes:** Verifies note metadata appears in the sidebar first, keeping content absent (lazy-loaded) until selected.
- **9. Lazy Loading Validation:** Confirms that clicking a placeholder note triggers a server request to fetch the actual content.
- **12. Delete Note Validation:** Deletes a note, verifying it is removed from the sidebar and backend database.
- **14. Offline Mode:** Simulates offline state. Creates note offline (saved locally), reconnects, and verifies it automatically syncs.
- **15. Multi Tab Validation:** Validates editing a note in one tab updates content in another tab without data loss.
- **16. Browser Refresh During Sync:** Refreshes browser mid-sync to verify no duplicate records are generated and no data is corrupted.
- **17. Rapid Note Switching:** Switches notes rapidly, validating that previous pending sync requests are cancelled and the correct note loads.
- **18. Rapid Switching While Editing:** Edits a note and switches away, verifying draft contents are preserved.
- **19. Large Note Validation:** Tests notepad limits with note sizes of 500KB, 900KB, 1MB, and 1.1MB, checking that size warnings are displayed.

5. Understanding the QA Test Reports

Every execution generates Word and/or PDF reports containing complete execution artifacts:

- **Execution Metadata:** Displays the test name, pass/fail status, date, browser, and duration.
- **Reproduction Steps:** Lists each step performed with a detailed explanation, time, and full-page screenshot.
- **Database Snapshot:** Parses the local browser IndexedDB database to display the internal records for verification.
- **Bug Ticket Logs:** For failures, compiles a red BUG DETECTED card documenting the expected vs actual result, screenshot, console logs, network requests, and stack trace.